A SOBER SPACE WITH A DIMENSION FUNCTION IS CATENARY, AND THE DIMENSION FUNCTION COMPUTES CODIMENSIONS OF CLOSURES

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Allowed standard prerequisites (editorial).

- The notion of a dimension function (an integer-valued function decreasing along specializations and respecting immediate specializations), and the catenary property as the equality of codimension sums in chains. Set up in the author's section on dimension functions, immediately before the counted statement. Fixes the hypothesis and the identity that the statement asserts.

Lemma 0.1. *Let X be a topological space. If X is sober and has a dimension function, then X is catenary. Moreover, for any $x \rightsquigarrow y$ we have*

$$\delta(x) - \delta(y) = \mathit{codim}\left(\overline{\{y\}},\ \overline{\{x\}}\right).$$

Proof. Suppose $Y \subset Y' \subset X$ are irreducible closed subsets. Let $\xi \in Y$, $\xi' \in Y'$ be their generic points. Then we see immediately from the definitions that $\mathrm{codim}(Y, Y') \leq \delta(\xi) - \delta(\xi') < \infty$. In fact the first inequality is an equality. Namely, suppose

$$Y = Y_0 \subset Y_1 \subset \ldots \subset Y_e = Y'$$

is any maximal chain of irreducible closed subsets. Let $\xi_i \in Y_i$ denote the generic point. Then we see that $\xi_i \rightsquigarrow \xi_{i+1}$ is an immediate specialization. Hence we see that $e = \delta(\xi) - \delta(\xi')$ as desired. This also proves the last statement of the lemma. □

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